Kowsar Mahmud Pappu

01961994513 | kowsarmahmud156@gmail.com | linkedin.com/in/kowsar-mahmud-pappu-9ba714194 | leetcode.com/Kowsar156 | github.com/Kowsar156

Education

Bangladesh University of Engineering & Technology — Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering (CGPA: 3.30) — *2019 – 2024*

St. Joseph Higher Secondary School — Dhaka, Bangladesh
Science (GPA: 5.00/5.00) — *2016 – 2018*

Nakhalpara Hossain Ali High School — Dhaka, Bangladesh
Science (GPA: 5.00/5.00) — *2012 – 2016*

Experience

Software Engineer I — June 2024 – Present
Synesis IT PLC — *Dhaka, Bangladesh*

- Developing & optimizing a scalable video conferencing web application named **Convay** as part of a government initiative with the Bangladesh Computer Council, funded by The World Bank
- Developed and maintained four ReactJS-based frontends aligned with microservice architecture, delivering scalable and high-performance user interfaces for the Convay platform
- Worked with `Consul` & `Cloud Gateway Frameworks` for a Gateway service that routes, filters & manages requests between client & multiple backend microservices & deployed/tested the gateway via a `Jenkins` pipeline working closely with the DevOps team
- Strengthened security of backend services by implementing JWT validation at Spring Cloud Gateway, resolving ∼90% of VAPT findings and satisfying security concerns of a major client
- Upgraded dependencies across all backend microservices - replaced `Zuul` with language specific `Cloud Gateway Framework`, and updated backend language and framework versions
- Designed and implemented an extensible provider-agnostic Payment Gateway Service for **Convay** using Java interfaces and service abstractions, allowing integration of providers like SSLCommerz (future-ready for Stripe/others)
- Leveraged AI-assisted development tools (ChatGPT, Cursor & Claude Code) to accelerate feature development, debug complex issues, and iteratively refine solutions through structured multi-step planning and agentic workflows

Machine Learning Intern — May 2023 – June 2023
North West Power Generation Company Limited — *Dhaka, Bangladesh*

- Used basic Machine Learning to make digital twins of Power Station machines and predict their lifecycles
- Datasets were supplied by Sirajganj Combined Cycle Power Plant and Sirajganj Grid Connected Solar Photovoltaic Power Plant authorities

Projects

Budget Buddy | *Flask, React, PostgreSQL* | **Code**

- Web based buddy that will help making financial decisions and keep the user up-to-date about financial events
- Bank account details, regular expenditure, saving goals and tax information tracking features
- JSON Web Token library was used for authentication & authorization purposes

Car Showroom System | *Java* | **Code**

- A simple car showroom management system made with Java
- Maintains car information, allows manufacturers to sell and common users to view, search and buy different types of cars
- OOP, Threading & Networking concepts were used

Simple Server System | *Java* | **Code**

- A simple server system made with Java
- Multiple clients can access the contents of the server simultaneously and upload/download files from/into local machines

Pneumothorax Detection Using Chest X-Ray Images | *Python, Pytorch* | **Code**

- Machine Learning project to detect pneumothorax using chest x-ray image
- Google Colaboratory was used to experiment with different Deep Learning architectures
- ResNet, EfficientNet and MaxViT architectures were explored

Technical Skills

Languages: C/C++, Java, Python, JavaScript, TypeScript, HTML
Frameworks and Database: ReactJS, ExpressJS, Spring Boot, Django, Flask, MySQL, PostgreSQL
Developer & Documentation Tools: Git & Github, VS Code, LaTeX, Google Colab
AI-Assisted Development: Regular use of ChatGPT, Cursor, and Claude Code for code generation, debugging, refactoring, and rapid prototyping; experienced with agentic workflows, multi-step planning, and iterative problem decomposition